

One Learned Operator, Three Audits: the Input–Architecture Curve, the Strengthened Comparator, and What the Network Actually Found

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Abstract

The $SU(2)$ spectroscopy chapter of this thesis rests on one number: a gauge-equivariant lattice transformer trained as a variational 0^{++} operator carries more ground-state weight than the optimal combination of the classical smeared basis, $\Delta A_0 = +0.078 \pm 0.022$ (3.6σ) over two independently sampled ensembles at a consistent mass. A single number of that shape invites three separate objections, and this report is the record of answering all three offline, on the same 400 held-out configurations per ensemble, with the estimator and the readings fixed in writing before the numbers existed. **Was the opponent strong enough?** Against the input-matched seven-level classical ladder the claim reads $+0.095 \pm 0.021$; against the strongest basis the repository builds — 21 operators, seven smearing levels times three loop shapes — it falls to $+0.038 \pm 0.027$, so genuine loop-shape variety closes about 60% of the gap and leaves a $1\text{--}2\sigma$ edge. Reading that arm at all required dropping the near-null directions of $C(t_0)$ rather than flooring them, and the pre-registered rule for choosing between the two turned out to have no answer, for a reason worth recording: the GEVP at $t_0 = 1$ maximises $C(2)/C(1)$ while the variational gate tests $C(2)/C(0)$. **Is it the architecture or just richer inputs?** Replacing the single number by A_0 as a function of the input content handed to *both* methods gives a curve on which the learned operator sits above the classical one at every resolvable point ($+0.252 \pm 0.056$ on thin links, $+0.075 \pm 0.019$ at four smearing levels), an untrained network of identical architecture sits below the classical one everywhere, and a width control shows the gain is input content and not capacity. None of the three pre-registered readings survives its own test unmodified. **What did the network find?** Because the equal-time connected correlator *is* the Hilbert-space inner product, the split $O_{\text{GELT}} = P + r$ into a classical projection and an orthogonal remainder is exact. The same trained operator is 21.1%, 12.8% and 2.5% outside the 4-, 7- and 21-operator spans, deleting r costs the entire advantage ($+0.097 \pm 0.021$ against the strong arm), and 80% of r is rectangular Wilson loops the network rediscovered. The decisive control is an untrained network: it is *further* outside the span (16–74%) than the trained one, so the norm fraction is architectural — what is learned is the amplitude share Z_r/Z_G (0.146 against 0.047), the fact that the residual becomes loop-shaped at all (80% against 0.3%), and the sign of $\Delta A_0(\text{net} - P)$, which flips to -0.091 ± 0.042 without training.

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1 The claim, and the three questions it invites

The result being audited is stated in `reports/glueball/glueball_spectroscopy.tex` and reproduced here in one sentence. On an anisotropic $SU(2)$ lattice ($12^3 \times 24$, $\beta = 2.4$, $\xi = 3.0$), a GELT network trained per timeslice on the Rayleigh-quotient loss $-C(1)/C(0)$ produces a single scalar operator whose ground-state overlap exceeds that of the variationally optimal combination of a four-level APE smearing basis:

$$\Delta A_0 = A_0(\text{GELT}) - A_0(\text{GEVP}) = +0.078 \pm 0.022 \quad (3.6\sigma), \quad (1)$$

combining two independently sampled ensembles, at a fitted mass consistent with unchanged.¹

Equation (1) is a comparison, and every comparison of this shape admits exactly three lines of attack. They are logically independent, they require different evidence, and this report devotes one section to each.

Was the opponent strong enough? The classical arm of Eq. (1) is four smearing levels of a *single* loop shape, the 1×1 spatial plaquette. A practitioner building a Morningstar–Peardon basis would use many shapes. If a stronger classical basis closes the gap, the result is about a weak comparator rather than about a network. Section 3.

Is it the architecture, or just richer inputs? The network is fed smeared link configurations, and one that merely *relays* its input advantage is not an interesting result. The replacement for a single ΔA_0 is a *curve*: A_0 as a function of the input content handed

¹`notes/glueball_spectroscopy.md`, “Replication results (2026-07-05)”. The two inputs are $+0.066 \pm 0.031$ on the Run-5 ensemble and $+0.089 \pm 0.030$ on the independent seed-1 ensemble.

to both methods, with a third trace — the same architecture, untrained — separating architecture from training. Section 4.

What did the network actually find? Granting the win, a referee asks whether the advantage is *new operator content* or the *same content in a combination the GEVP failed to locate*. The first says the classical variational space has a ceiling; the second says the estimator was noisy and more configurations would fix it. The question has an exact answer, because operators acting on the vacuum are vectors in a Hilbert space whose inner product the correlator already computes. Section 5.

All three audits are pure re-analyses of arrays that already exist: the test-split time series $\bar{O}(t)$ of every operator, learned and classical, on the 400 held-out configurations of each ensemble. Every number here was produced on a laptop CPU in seconds to minutes from files tracked in `dumps/` — no GPU time, no training, and, importantly for the first two audits, no possibility of quietly re-training until the answer improved.

The pre-registration. Sections 1 and 2 of `notes/fable5.1_10-09_audit.md` fix the observable, the estimator, the fit conventions, the grid of measurement points, and three named readings with their operational tests, and were committed *before* any of the numbers below existed. This is the only defence against the objection that a curve with five points and three traces offers many places to stop looking. It is also why this report says twice, in Sections 3.2 and 4.3, that a pre-registered rule failed on contact with the data — recording the amendment and its reason is the cost of the discipline, and the alternative (silently choosing what works) is exactly what the discipline exists to prevent.

2 Method

This section is the minimum needed to read Sections 3 to 5. The path integral, the transfer matrix, the spectral decomposition of a Euclidean correlator, APE smearing, the anisotropic action, and the heat-bath/overrelaxation sampler are developed from first principles in the companion report `glueball_spectroscopy.tex` and are not repeated.

2.1 Correlators, the cosh fit, and the overlap fraction

Let $\bar{O}(t)$ be the zero-momentum, timeslice-summed value of a scalar operator on one gauge configuration. The 0^{++} channel has a nonzero vacuum expectation value, so the object with a spectral decomposition is the *connected* correlator

$$C(\Delta) = \langle d(t) d(t + \Delta) \rangle, \quad d(t) = \bar{O}(t) - \langle \bar{O} \rangle, \quad (2)$$

averaged over t and over configurations. On a periodic lattice of temporal extent N_t the transfer matrix gives

$$C(\Delta) = \sum_{n \geq 1} |Z_n|^2 \left[e^{-m_n \Delta} + e^{-m_n (N_t - \Delta)} \right], \quad Z_n = \langle 0 | \bar{O} | n \rangle. \quad (3)$$

Fitting the single-state form $C(\Delta) \approx A [e^{-m \Delta} + e^{-m (N_t - \Delta)}]$ over a window in which the excited states have died gives (m, A) ; the mass m is the physics and $A = |Z_0|^2$ is the operator’s coupling to the ground state. The figure of merit for an *operator*, as opposed to a mass, is the fraction of the operator’s total norm that sits on the ground state,

$$A_0 = \frac{A (1 + e^{-m N_t})}{C(0)} \quad (4)$$

— the numerator is the ground-state contribution to $C(0)$ evaluated from the fit, the denominator the measured $C(0)$, which by Eq. (3) contains every state. A pure ground-state interpolator has $A_0 = 1$; excited-state contamination pushes it down.

Three properties of A_0 matter for what follows.

1. **It is the right figure of merit at fixed mass.** Two operators that agree on m but differ on A_0 measure the same physics with different amounts of contamination — the comparison this report needs, and Morningstar and Peardon’s own operator-quality metric. It is also *exactly what the training loss maximises*: the Rayleigh quotient $-C(1)/C(0)$ is minimised by the operator of largest ground-state fraction at the mass gap, so objective and metric coincide.
2. **$A_0 > 1$ can occur, and is a fit artifact rather than a probability violation.** A_0 is a ratio of a *fitted* amplitude to a *measured* $C(0)$, estimated on a finite sample; when the operator is nearly pure, the two estimates differ by noise in both directions. The seed-1 ensemble reads $A_0 = 1.013 \pm 0.062$ for the four-level learned operator and 1.066 ± 0.059 for the seven-level one. These are quoted unclipped, with $1 - A_0$ carrying its sign; clipping at one would bias every difference in which they appear.
3. **It saturates.** $A_0 \leq 1$, and every strong arm on these ensembles already reads 0.90–1.07. Two methods separated by less than the errors near the ceiling are not thereby shown to be equivalent; the observable has simply run out of range. This is pre-registered as the *saturation clause* and is enforced mechanically in Section 4.3.

Fit and error conventions. (m, A) come from `fit_cosh_correlator`: a profiled- A grid scan with parabolic refine, dependency-free so that it is safe to call inside every jackknife replica, with diagonal χ^2 weights σ_Δ fixed from the full sample. Errors are delete-block jackknife over configurations with block 10, and the *entire* fit — including, for the classical arm, the re-resolution of the GEVP eigenvector — is redone inside every replica. Differences between two operators are always *correlated* differences on shared configurations: the difference is formed inside each replica and jackknifed, not assembled from two independent error bars.

2.2 The classical comparator: the GEVP ground vector

A learned operator is one operator. Comparing it against a *basis* would be unfair in the network’s favour; comparing it against the best *single* member of that basis would be unfair against it. The apples-to-apples object is the optimal single operator inside the basis’s span, and the variational method supplies it. Given the correlator matrix $C_{ab}(\Delta)$ of a basis $\{\bar{O}_a\}$, solve

$$C(t_d)v = \lambda C(t_0)v, \quad (5)$$

take v_0 the eigenvector of the largest λ , and define the *projected operator* $\bar{O}_{v_0} = \sum_a (v_0)_a \bar{O}_a$. This is the combination of basis members that maximises the decay ratio $C(t_d)/C(t_0)$; it is what `gevp_ground_vector` returns and what every “classical” A_0 in this report is measured on. Throughout, $t_0 = 1$ and $t_d = 2$.

2.3 The whitening problem: flooring versus truncation

Equation (5) is solved by whitening: diagonalise $C(t_0) = QsQ^\top$, form $W = Qs^{-1/2}$, and take the top eigenvector of $W^\top C(t_d)W$. When the basis is a long ladder of near-degenerate operators — seven smearing levels of the same plaquette are almost collinear, and 21 operators built as levels $\times$ shapes are worse — $C(t_0)$ estimated on 400 configurations is numerically singular and can be indefinite. The 21-operator arm reads $\text{cond } C(t_0) = \infty$ on both ensembles. There are two repairs:

Flooring. Clamp $s \rightarrow \max(s, \epsilon s_{\max})$ and keep all n_{ops} directions. Every direction survives, including ones whose scale is set entirely by ϵ .

Truncation. Keep only the k directions with $s > \epsilon s_{\max}$ and *drop* the rest, so the GEVP is solved in a k -dimensional subspace with $k \leq n_{\text{ops}}$.

The difference is not cosmetic. A floored near-null direction is a direction whose normalisation is arbitrary, and the whitened problem can and did select one: an early run of this audit reported the strengthened arms as *worse* than the basis they contain — $A_0 = 0.072$ for the 21-operator arm against 0.837 for its own 4-operator subset — and duly announced that the claim survived at 8.8σ .² Since **published** is a strict *subset* of the others, their GEVP optimises over a larger span and cannot interpolate worse; the violation of that superset property is what exposed the estimator. It is now a hard gate in the script.

Because the choice of repair can move a number, it had to be made by a rule fixed in advance rather than by which setting gave the nicer answer. That rule, its failure and its amendment are Section 3.2, and the evidence is the four-setting table there.

2.4 The decomposition: an inner product, not a fit

The third audit rests on an observation that is exact rather than approximate. Operators acting on the vacuum are vectors in a Hilbert space, and the equal-time connected correlator *is* the inner product on that space:

$$\langle O_a, O_b \rangle \equiv C_{ab}(0) = \langle d_a(t) d_b(t) \rangle, \quad (6)$$

which is what `connected_correlator_matrix` already returns at $\Delta = 0$. Consequently “the part of the learned operator that the classical basis cannot express” is a *literal orthogonal projection*:

$$O_{\text{GELT}} = P + r, \quad P = \text{proj}_{\text{span}\{\bar{O}_a\}} O_{\text{GELT}}, \quad r \perp \text{span}\{\bar{O}_a\}. \quad (7)$$

Nothing here is fitted to come out any particular way. Three consequences make the split worth measuring.

1. **P is the strongest possible classical opponent.** By construction it is the best approximation to O_{GELT} *inside the span* — better than the GEVP vector, better than the most-smeared member, better than any classical operator chosen for any other reason. So $\Delta A_0(\text{GELT} - P) > 0$ is a *lower bound* on the advantage over the entire span, and it is immune to the objection that answers Section 3 the hard way: one cannot claim the comparator was badly chosen when the comparator is the learned operator’s own shadow.
2. **The difference is better conditioned than the published one.** Equation (1) compares GELT to the GEVP vector, a *different* operator correlated with GELT only through shared configurations. P is GELT’s own projection, so the two share every fluctuation living in the span and much of that variance cancels.
3. **The same statistic gives the missing scale.** “12.9% of the norm² lies outside a four-dimensional span” means nothing alone: any operator outside a 4D subspace has a large orthogonal component. The scale is the basis’s *own* increments — how much new content each rung of the smearing ladder adds to the span below it — which costs no new data and says whether the ladder is still growing.

²`notes/audit_2026-09-06.md` §6. The failure mode is instructive precisely because it pointed the flattering way.

The amplitude split, at one fixed mass. Z is linear in the operator, so $Z_G = Z_P + Z_r$ holds *exactly* — but only if all three are read against the same exponential. Letting each piece fit its own mass destroys the additivity that makes the split interpretable, so amplitudes come from a least-squares fit of $C(\Delta) \approx A \cosh_{\text{ref}}(\Delta)$ at the learned operator’s reference mass on that ensemble. The share Z_r/Z_G is then read off the *cross*-correlator, $C_{Gr}(\Delta) \rightarrow Z_G Z_r e^{-m\Delta}$ against $C_{GG}(\Delta) \rightarrow Z_G^2 e^{-m\Delta}$, which carries its own sign and needs no square root. For a network whose own cosh fit does not converge — an untrained one — the reference mass is passed in from the trained net on the same ensemble (`--m-ref`).

2.5 Conventions, all pre-registered

Table 1 is the complete list. It was fixed in `notes/fable5.1_10-09_audit.md` §1.4 before the measurements, with the single exception noted in its last row and explained in Section 3.2.

Table 1: Fixed conventions. Every number in this report is measured under these, on both ensembles, with no per-point tuning.

item	value	where
fit window	$\Delta \in [2, 7]$, cosh, profiled- A grid	<code>fit_glueball_overlap.py</code>
jackknife	delete-block, block 10, whole fit per replica	same
GEVP	$t_0 = 1$, $t_d = 2$, v_0 -projected operator	<code>gevp_ground_vector</code>
whitening	truncate , $\epsilon = 10^{-4}$ relative	adopted, §3.2
test split	contiguous last 20% (400 configs)	<code>train_glueball.py</code>
ensembles	run5 (seed 0), ens1 (seed 1), inverse-variance	§1
lattice	$SU(2)$, $12^3 \times 24$, $\beta = 2.4$, $\xi = 3.0$	—
width policy	every new trained point at $d_{\text{model}} = 24$	§4.4
random trace	init seeds 0, 1, 2; error is the larger of seed spread and mean jackknife error	§4.5

The two ensembles are sampled from scratch with different sampler seeds and combined inverse-variance. They are not two halves of one ensemble: run5 and ens1 have different classical anchors ($m_{\text{eff}}(\Delta = 1) \approx 0.357$ and 0.390) and different Rayleigh-loss floors, so a result replicating across them has replicated across the ensemble noise, the dominant systematic here.

3 The fair fight: was the comparator a straw man?

3.1 The arms

Six classical bases are built on the same 400 test configurations, projected by the same `gevp_ground_vector`, fitted on the same window with the same fixed σ_Δ and jackknifed on the same blocks — the protocol is *imported* from `fit_glueball_overlap.py`, so “same conventions as the published table” is a fact about the call graph. Each basis is a set of spatial Wilson loops of shapes $R \times T$ built from links smeared with n cumulative APE steps:

arm	content	n_{ops}	role
thin	$n \in \{0\}$, shape 1×1	1	the unsmeared plaquette; the curve’s left end
published	$n \in \{0, 2, 4, 6\}$, shape 1×1	4	the arm of Eq. (1)
deep	$n \in \{0, 2, 4, 6, 8, 12, 16\}$, shape 1×1	7	more radius; input-matched to the 7-level net
shapes	$n \in \{0\}$, five $R \times T$ shapes	5	extent without smearing
shapes_sm	$n \in \{0, 2, 4, 6\}$, five shapes	20	levels $\times$ shapes, shallow ladder
full	$n \in \{0, 2, 4, 6, 8, 12, 16\}$, three shapes	21	the strongest basis available

The loop shapes are cubic-symmetrised, so each “shape” is already a sum over the orientations the 0^{++} channel requires. **deep** is the *input-matched* comparator: its smearing levels are exactly

the channels the seven-level network was fed, so a win there cannot be explained by the network having seen more smearing than the classical side. Two gates run before any verdict: the *slice gate* (the cached basis is measured on the same configurations as the dump — a failure means wrong configurations, not physics) and the *superset gate* (no superset arm reads below published by more than 2σ , which by Section 2.3 cannot happen for a correct estimator, so a violation is a bug report and not a result).

3.2 The estimator table, and a pre-registered rule with no answer

The rule fixed in advance was: run every arm on both seven-level dumps under (a) floor at $\epsilon = 10^{-4}$, (b) truncate at $\epsilon = 10^{-4}$, (c) (b) plus collinearity pruning at $\rho = 0.999$, (d) (b) plus pruning at $\rho = 0.99$; then quote the *least invasive setting under which every arm passes both gates on both ensembles and **published** is unchanged to three decimals*. Sixteen offline runs, 45 seconds in total on a laptop, produced Table 2.

Table 2: Classical A_0 per arm under the four estimator settings, both ensembles. “kept” is the number of $C(t_0)$ directions above $10^{-4}s_{\max}$ under (b); “prune” the number of operators left by (c)/(d) before truncation. Setting (c) reads (b) to 0.001 everywhere and is omitted from the A_0 columns.

arm	n_{ops}	kept (b)	prune (c/d)	A_0 run5: (a) $\rightarrow$ (b) $\rightarrow$ (d)	A_0 ens1: (a) $\rightarrow$ (b) $\rightarrow$ (d)
thin	1	1	1 / 1	0.390(60)	0.484(128)
published	4	4	4 / 4	0.837(56), unchanged	0.925(71), unchanged
deep	7	5	7 / 7	0.903 $\rightarrow$ 0.905 $\rightarrow$ 0.905	0.939 $\rightarrow$ 0.939 $\rightarrow$ 0.939
shapes	5	5	5 / 5	0.500(110) FELL BACK	0.612(110) FELL BACK
shapes_sm	20	11	20 / 16	0.934 $\rightarrow$ 0.913 $\rightarrow$ 0.910	1.003 $\rightarrow$ 1.004 $\rightarrow$ 1.004 FB
full	21	9	20 / 14	0.942 $\rightarrow$ 0.938 $\rightarrow$ 0.936	0.960 $\rightarrow$ 0.960 $\rightarrow$ 0.978

Table 3: Combined $\Delta A_0 = \text{GELT} - \text{arm} (\text{run5} \oplus \text{ens1}, \text{inverse variance})$ under the four settings, for both trained networks. In brackets, the run5 – ens1 tension. “same” means identical to the quoted digits.

arm	(a) floor	(b) truncate	(c) + prune 0.999	(d) + prune 0.99
<i>seven-level GELT</i>				
thin	$+0.569 \pm 0.069$	same	same	same
published	$+0.131 \pm 0.029$ (0.4 σ)	same	same	same
deep	$+0.097 \pm 0.021$ (1.7 σ)	$+0.095 \pm 0.021$ (1.8 σ)	$+0.095 \pm 0.021$	$+0.095 \pm 0.021$
shapes	$+0.454 \pm 0.084$	same	same	same
shapes_sm	$+0.058 \pm 0.016$ (0.7 σ)	$+0.059 \pm 0.029$ (0.2 σ)	$+0.059 \pm 0.029$	$+0.060 \pm 0.026$
full	$+0.032 \pm 0.026$ (1.5 σ)	$+\mathbf{0.038} \pm \mathbf{0.027}$ (1.4 σ)	$+0.037 \pm 0.026$	$+0.042 \pm 0.022$
<i>four-level GELT</i>				
thin	$+0.517 \pm 0.063$	same	same	same
published	$+0.077 \pm 0.022$ (0.5 σ)	same	same	same
deep	$+0.041 \pm 0.025$ (1.5 σ)	$+0.039 \pm 0.025$	$+0.039 \pm 0.025$	$+0.039 \pm 0.025$
shapes_sm	$+0.002 \pm 0.024$	$+0.006 \pm 0.035$	$+0.006 \pm 0.035$	$+0.008 \pm 0.032$
full	-0.022 ± 0.026 (1.4 σ)	-0.013 ± 0.030	-0.013 ± 0.030	-0.011 ± 0.023

The gates pass. published reproduces $+0.066/+0.089$ against the four-level networks — i.e. Eq. (1) — and its A_0 is unchanged to the third decimal under all four settings. deep reproduces $+0.052/+0.127$ under (a) and reads $+0.050/+0.127$ under (b)–(d): truncation drops 2 of its 7 directions and moves the answer by 0.06σ . The superset gate passes everywhere.

The rule has no answer as written. Two arms, `shapes` and `shapes_sm`, fall back to their best single member on `ens1` under *all four* settings, so no setting satisfies “every arm passes on both ensembles”. The interesting part is why, because the natural diagnosis — ill-conditioning — is wrong: `shapes` falls back at $\text{cond } C(t_0) = 87$, which is a perfectly healthy matrix.

Checked directly on the `ens1` cache, the v_0 -projected `shapes` operator *beats* its best member on $C(2)/C(1)$, 0.640 against 0.597, and *loses* on $C(2)/C(0)$, 0.092 against 0.206. Both are correct and they do not conflict: the GEVP at $(t_0, t_d) = (1, 2)$ is by construction the maximiser of $C(2)/C(1)$ and says nothing whatever about $C(2)/C(0)$ — while the variational gate, and A_0 itself, are statements about $C(\cdot)/C(0)$. The projection bought its improved $1 \rightarrow 2$ decay at the cost of a large $\Delta = 0$ piece ($C(1)/C(0) = 0.14$): GEVP-optimal and a poor A_0 interpolator at once. `shapes_sm` on `ens1` is the same story ($C(2)/C(1)$: 0.720 vs 0.712; $C(2)/C(0)$: 0.430 vs 0.490).

The consequence is structural, not numerical. The variational gate is a theorem only at $t_0 = 0$, where the whitening metric and the normalisation of A_0 are the same matrix; at $t_0 = 1$ they are different matrices, so no change to the *whitening* can rescue an arm that fails this way. A rule no setting in the family can satisfy cannot select a setting.

The amendment, and why it moves nothing. The rule was therefore amended, and recorded as a deviation: it now ranges over the arms on the curve’s main chain (`thin` $\subset$ `published` $\subset$ `deep` $\subset$ `full`), since `shapes` is not on the curve at all and `shapes_sm` is the off-chain point, reported with its `ens1` fallback flagged rather than dropped. Among the settings that pass for those arms, the least invasive one that removes `full`’s null directions is (b): truncate at $\epsilon = 10^{-4}$, no pruning — the estimator for every number in this report. The amendment changes no conclusion, and Table 3 is the evidence rather than an assurance: every combined value there is stable across (a)–(d) to ≤ 0.01 , well inside its own error, and `published` is identical to three decimals under all four.

3.3 The verdict on loop shapes

With truncation, `full` becomes readable: a nine-dimensional GEVP that passes both gates on both ensembles, whose A_0 barely moves from the floor path ($0.942 \rightarrow 0.938$ on `run5`, $0.960 \rightarrow 0.960$ on `ens1`). This answers the question the earlier audit left explicitly open.

$$\Delta A_0(\text{full}) = +0.038 \pm 0.027 \text{ (1.4}\sigma\text{)} \quad \text{against} \quad \Delta A_0(\text{deep}) = +0.095 \pm 0.021 \text{ (4.5}\sigma\text{)}. \quad (8)$$

Loop shapes stacked on top of the deep ladder close about 60% of the gap, and what is left is a $1\text{--}2\sigma$ edge in the network’s favour. That is precisely the pre-registered expectation — “parity within a few hundredths or a $1\text{--}2\sigma$ GELT edge” — which is worth saying because it is one of the few predictions in this block that landed where it was aimed. The four-level network, whose inputs contain neither the deep levels nor any shapes, is at parity with `full` (-0.013 ± 0.030).

Two earlier statements that do not survive. The first pass of this audit, written when `full` was unreadable, recorded two claims that the truncated estimator refutes.³

- “*full*’s ΔA_0 is not reproducible between ensembles — an $8\times$ swing.” The two values, $+0.013 \pm 0.029$ and $+0.105 \pm 0.056$, differ by 1.4σ . The $8\times$ is the ratio of two small central values, which is not a measure of tension.

³`notes/audit_2026-09-06.md` §6.5, superseded on both points by `notes/fable5.1_10-09_audit.md` §8.1.

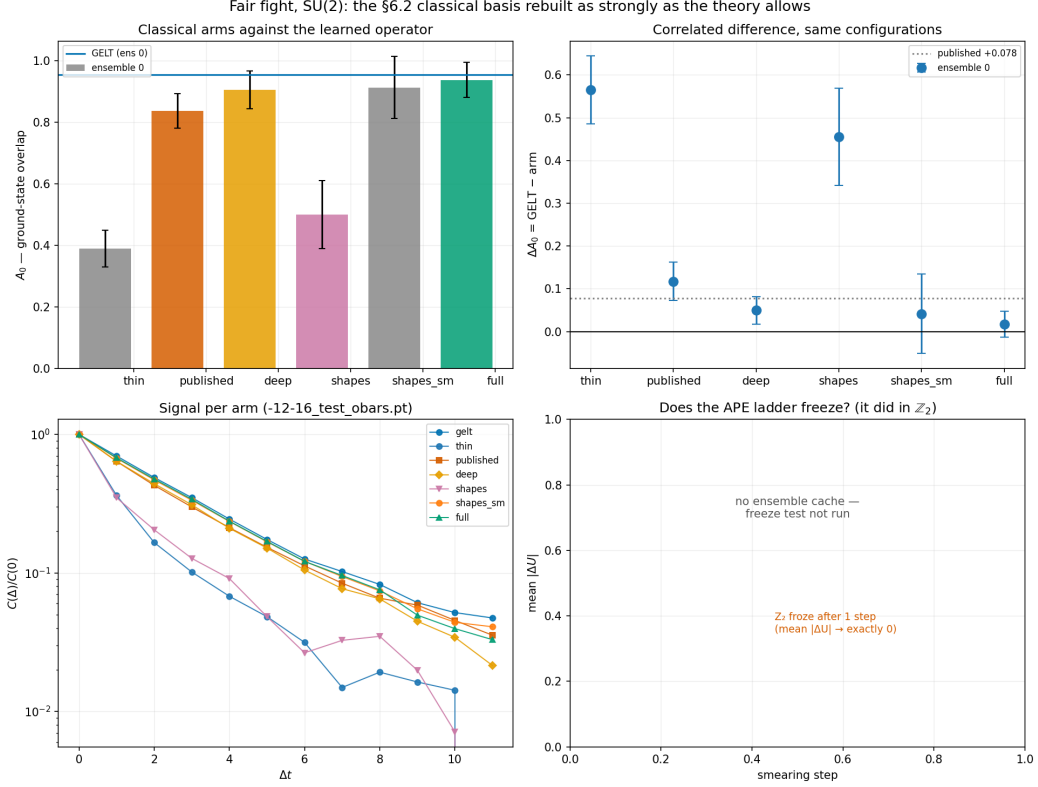


Figure 1: The fair fight on the run5 ensemble, seven-level network, adopted (truncated) estimator. Top left: A_0 of each classical arm against the learned operator (horizontal line). Top right: the correlated difference ΔA_0 per arm on the same configurations, with the published +0.078 marked. Bottom left: $C(\Delta)/C(0)$ per arm — **thin** and **shapes** visibly lose their signal by $\Delta \approx 5$, which is why their A_0 is poor and why neither is a serious opponent. The bottom-right panel is the $\mathbb{Z}_2$ smearing-freeze diagnostic, inactive here because this run reads the cached basis rather than re-smearing an ensemble. Regenerate with `SFF_TRUNCATE=1 SFF_BASES=1 python scripts/su2_fair_fight.py dumps/best_glueball_gelt_sm0-2-4-6-8-12-16_test_obars.pt`.

- “**full**’s eigenvector is fixed by the ϵ floor, not by signal.” Dropping the floored directions entirely moves **full**’s A_0 by 0.004. The floor was not steering the answer. What the earlier note got right is that $\text{cond} = \infty$ made that impossible to know without this check — the objection was sound, the conclusion was not.

What this does and does not license. *Does:* the published comparator was under-powered, and the honest headline against an input-matched classical basis is $+0.095 \pm 0.021$ rather than $+0.078 \pm 0.022$; a classical basis with real loop-shape variety closes most of the remaining gap, and the network retains a $1-2\sigma$ edge over 21 hand-built operators. *Does not:* treat the 21-operator comparison as a measurement of equal quality to the others. **deep** and **full** both sit at $A_0 \geq 0.90$ on both ensembles, inside the saturation band of Section 2.1, so Eq. (8) is read off a compressed observable and the 60% is a rough fraction, not a measured ratio with an error. Extending past that ceiling needs a harder observable, not more configurations (Section 7).

4 The curve: architecture or inputs?

4.1 Definition

The fair fight strengthens the opponent while holding the network fixed. The curve does something different: it varies the *input content* x handed to *both* methods and asks how the difference behaves. Write $A_0(x)$ for

$$x \in \{ \text{thin}, 4\text{lv}, 7\text{lv}, 4\text{lv}+5\text{sh}, 7\text{lv}+3\text{sh} \}, \quad (9)$$

ordered by inclusion along the main chain $\text{thin} \subset 4\text{lv} \subset 7\text{lv} \subset 7\text{lv}+3\text{sh}$, with $4\text{lv}+5\text{sh}$ an off-chain point (neither a subset nor a superset of 7lv). Three traces are measured at each x :

trace	what it is	source
classical GEVP(x)	the v_0 -projected GEVP at content x	the arm of that name
trained GELT(x)	a GELT trained on channels with content x	<code>train_glueball.py</code>
random GELT(x)	the same architecture, untrained, same inputs	eval-only, three init seeds

Every point is measured on the same 400 held-out configurations, every difference at fixed x is a correlated blocked jackknife on shared configurations, and both ensembles are combined inverse-variance. The third trace is the load-bearing one for this section’s title: if the architecture alone, at random initialisation, already beat the classical GEVP of the same inputs, “learned” would be the wrong word for what the trained network does.

4.2 The measurements

Twenty-six dumps, one fair-fight run each under the adopted estimator (73 seconds on a laptop for all of them), assembled by `scripts/input_architecture_curve.py`, which fits nothing and re-estimates nothing — the numbers in Table 4 are the numbers `su2_fair_fight.py` printed.

Table 4: A_0 against input content. “classical” is the GEVP over the same input content; “trained” is at $d_{\text{model}} = 24$; “random” is the mean over init seeds 0,1,2 under the error rule of Table 1. ΔA_0 is the correlated trained–classical difference. The last column is the rung-equivalence entry $\Delta A_0(\text{GELT}(x_k) - \text{GEVP}(x_{k+1}))$: the learned operator at one input depth against the classical basis at the next. Generated verbatim as `results/fair_fight/input_architecture_curve.tex`.

x	ens	A_0 classical	A_0 trained	A_0 random	ΔA_0	combined	rung ΔA_0
thin	run5	0.390(60)	0.651(77)	0.249(62)	$+0.261 \pm 0.063$	$+0.252 \pm 0.056$	-0.201 ± 0.048
	ens1	0.484(128)	0.706(73)	0.280(109)	$+0.221 \pm 0.117$		
4lv	run5	0.837(56)	0.911(51)	0.497(175)	$+0.073 \pm 0.026$	$+0.075 \pm 0.019$	$+0.038 \pm 0.024$
	ens1	0.925(71)	1.002(61)	0.532(198)	$+0.078 \pm 0.029$		
7lv	run5	0.905(62)	0.955(57)	0.730(343)	$+0.050 \pm 0.033$	$+0.095 \pm 0.021$	
	ens1	0.939(65)	1.066(59)	0.512(148)	$+0.127 \pm 0.027$		

Combined over ensembles, the classical trace reads 0.407 ± 0.054 , 0.870 ± 0.044 , 0.921 ± 0.045 ; the trained trace 0.680 ± 0.053 , 0.949 ± 0.039 , 1.008 ± 0.041 ; the random trace 0.257 ± 0.054 , 0.512 ± 0.131 , 0.547 ± 0.136 .

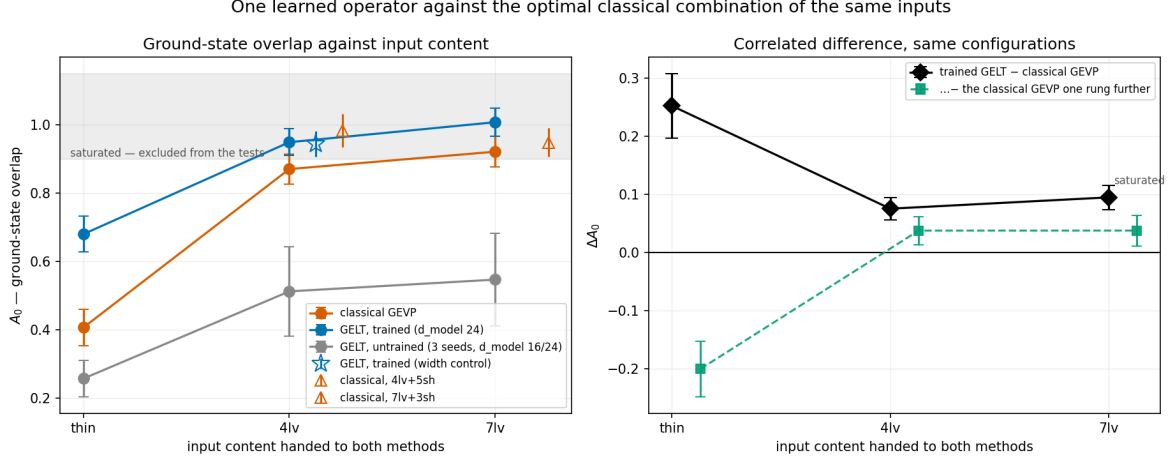


Figure 2: The input–architecture curve. Left: A_0 against input content for the three traces, ensembles inverse-variance combined, with the two off-chain classical points (`shapes_sm`, `full`) as open triangles and the $A_0 \geq 0.90$ saturation band shaded. The open star is the width control (Section 4.4). Right: the correlated difference $\Delta A_0(x)$, and the rung-equivalence difference $\text{GELT}(x_k) - \text{GEVP}(x_{k+1})$, which is the quantity asking whether the architecture is worth a smearing rung. Note the shape of the black curve: the advantage is large where the classical basis is poor and shrinks as it is fed.

4.3 The three readings, evaluated by their own tests

A point is *resolvable* when the combined classical $A_0 < 0.90$; above that the observable has saturated and cannot discriminate. This excludes 7lv ($A_0 = 0.921$) and leaves thin (0.407) and 4lv (0.870). The three readings and their pre-registered operational tests were fixed before any of Table 4 existed, and the script evaluates them mechanically.

R1 — the architecture buys input depth. *Test:* $\Delta A_0(x) > 2\sigma$ at every resolvable x , and rung-equivalence $A_0^{\text{GELT}}(x_k) \geq A_0^{\text{GEVP}}(x_{k+1})$ within errors for $k = \text{thin} \rightarrow 4\text{lv}$ and $4\text{lv} \rightarrow 7\text{lv}$. *Result:* *splits*. The first clause holds cleanly. The rung clause holds for $4\text{lv} \rightarrow 7\text{lv}$ ($+0.038 \pm 0.024$: a four-level GELT matches the seven-level classical basis) and **fails** for $\text{thin} \rightarrow 4\text{lv}$ (-0.201 ± 0.048 : a thin-input GELT does not reach the four-level basis). R1 does not hold as stated.

R2 — an efficient shortcut, not an extension. *Test:* $\Delta A_0(x)$ consistent with zero at some resolvable x where the GEVP still has $1 - A_0$ more than 3σ from zero. *Result:* *fails*. The traces never meet before saturation.

R3 — an additive advantage. *Test:* $\Delta A_0(x)$ consistent with a constant $c > 0$ over the resolvable points, $\chi^2/\text{dof} < 2$ and $c > 3\sigma$. *Result:* *fails*, and informatively. The constant fit gives $c = +0.094 \pm 0.018$ with $\chi^2/\text{dof} = 9.1$: the advantage is not additive, it *shrinks* as the inputs get richer, from $+0.25$ at thin links to $+0.08$ at four smearing levels.

That none of the three survives unmodified is the honest outcome and is more informative than a clean pass would have been. What the data say is a fourth statement the pre-registration did not name: the advantage is real at every resolvable input depth, it is largest where the classical side is weakest, and it decays toward the saturation ceiling rather than persisting as a fixed offset. The rung clause splits along the same axis — worth a whole smearing rung at four levels, not worth one at thin links — which is what one expects if the network’s contribution and the classical ladder’s contribution overlap increasingly as the ladder grows. Section 5 measures

that overlap directly and finds exactly that: $21.1\% \rightarrow 12.8\% \rightarrow 2.5\%$ of the learned operator lies outside the 4-, 7- and 21-operator spans.

4.4 The width control

The seven-level network is wider than the four-level one (d_{model} 24 against 16), because it was trained with more input channels. A curve drawn through both would then confound *input content* with *capacity* — the objection this section exists to answer, reappearing one level down. The control is one extra training: the four-level input set at $d_{\text{model}} = 24$, on both ensembles.

4lv trained network	run5	ens1
$d_{\text{model}} = 16$ (the published Run-5 net)	0.903(47)	1.013(62)
$d_{\text{model}} = 24$ (the curve’s point)	0.911(51)	1.002(61)

Width $16 \rightarrow 24$ changes nothing on either ensemble. The curve is therefore drawn at $d_{\text{model}} = 24$ throughout, so that moving along x changes the input content and nothing else, and the published $d_{\text{model}} = 16$ network becomes the control rather than a point. The 4lv $\rightarrow$ 7lv gain is input content, not capacity.

4.5 The random trace

The pre-registered expectation was $A_0^{\text{random}}(x) \leq A_0^{\text{GEVP}}(x)$ at every x , with the explicit note that a violation anywhere would mean the architecture alone beats the optimal linear combination of its own inputs, and that the “learned” attribution at that x would then rest on trained – random only.

It holds comfortably at all three points: 0.257 against 0.407 at thin links, 0.512 against 0.870 at four levels, 0.547 against 0.921 at seven. An untrained GELT is a legitimate gauge-invariant operator — it has a mass and a correlator — but a distinctly poor one, well below the classical basis built from the same configurations, so the advantage in Table 4 is attributable to training rather than to the architecture alone. This is also the first of the two controls `notes/operator_decomposition.md` §5 asked for, delivered at three input depths rather than one.

What this does and does not license. *Does:* “at every input depth this observable can resolve, one learned operator carries more ground-state weight than the optimal linear combination of the *same* inputs — $+0.252 \pm 0.056$ on thin links, $+0.075 \pm 0.019$ at four smearing levels; the advantage shrinks as the classical side is fed more, and at four levels it is worth a whole extra smearing rung while at thin links it is not.” *Does not:* extend past 7lv. Above four levels the observable saturates and cannot separate the methods further, and no amount of statistics changes that, because the ceiling is $A_0 = 1$ and not a noise level. Nor does it license a statement about the right-hand end of the chain: the shape-input network was never trained (Section 6), so 7lv+3sh compares a GELT *without* loop shapes to a GEVP *with* them, which is a fair reading only as long as the figure says so — it does, with that point drawn as an off-chain classical marker and no trained counterpart.

5 The decomposition: what did the network find?

Sections 3 and 4 establish that the learned operator wins, and against what. Neither says what it *is*. This section does, using the exact orthogonal split of Section 2.4, and it is where the report spends most of its numbers.

5.1 The gate, and why it belongs in the report

Before anything else is read, `operator_decomposition.py` recomputes the published comparison from *its own* code path and prints it beside the published values. It reproduces them to the last digit: $A_0(\text{GELT}) = 0.9029 \pm 0.0468$ against the GEVP’s 0.8374 ± 0.0555 , $\Delta A_0 = +0.0656 \pm 0.0306$ on run5 and $+0.0885 \pm 0.0302$ on ens1, 12.89(41)% and 11.69(30)% of the norm^2 outside the span, and a combined $\Delta A_0(\text{GELT} - P) = +0.0761 \pm 0.0192$ (4.0σ). A second gate came free when the span became selectable: with `--basis=...:deep` the GEVP arm reads $A_0 = 0.9053 / 0.9390$ and $\Delta A_0(\text{GELT} - \text{GEVP}) = +0.0496 / +0.1266$ — Section 3’s truncated `deep` row (0.905/0.939, +0.050/ +0.127), reproduced from a script that shares no estimator code with `su2_fair_fight.py` beyond the library fit routine.

This belongs in the report and not only in the code, because it is the only thing standing between a re-analysis and a silent discrepancy. The decomposition runs on the same dumps as the fair fight but through different projection, whitening and amplitude machinery; a disagreement of a few hundredths would be invisible in a table of decomposition results alone, and every conclusion below would inherit it. A gate that reproduces two independent published rows before printing anything new converts that risk into a precondition, at the cost of one extra column of an array the script already has.

5.2 The main table: one operator against three nested spans

The seven-level trained network is decomposed against three classical spans of increasing size, combined over both ensembles. The columns are the three statistics the split produces.

Table 5: The *same* trained seven-level operator against three nested classical spans, both ensembles combined inverse-variance. `full` is at the script default `--proj-eps = 10-12`; the last row repeats it at 10^{-4} , which retains 10 of the 21 Gram directions.

span	n_{ops}	norm^2 outside	Z_r/Z_G	$\Delta A_0(\text{GELT} - P)$
published (4lv)	4	0.2114 ± 0.0040	0.2293 ± 0.0094	$+0.157 \pm 0.027$ (5.8σ)
deep (7lv)	7	0.1277 ± 0.0025	0.1459 ± 0.0071	$+0.097 \pm 0.021$ (4.5σ)
full (7lv+3sh)	21	0.0251 ± 0.0005	0.0128 ± 0.0038	$+0.021 \pm 0.009$ (2.3σ)
full , $\epsilon = 10^{-4}$	$21 \rightarrow 10$	0.0452 ± 0.0010	0.0374 ± 0.0053	$+0.039 \pm 0.013$ (2.9σ)

Table 6: Per-ensemble detail against `deep`, the input-matched span. The last two rows are the statements that matter: dropping r costs the advantage, and it does so at unchanged mass.

against span{ deep }	run5	ens1
norm^2 fraction outside the span	0.1273 ± 0.0037	0.1279 ± 0.0033
Z_r/Z_G	0.1287 ± 0.0121	0.1550 ± 0.0088
$A_0(P)$ — best classical approximation of GELT	0.8836 ± 0.0578	0.9444 ± 0.0662
$A_0(r)$ — the remainder on its own	0.4612 ± 0.0697	0.4165 ± 0.0584
$\Delta A_0(\text{GELT} - P)$, correlated	$+0.0712 \pm 0.0304$	$+0.1212 \pm 0.0301$
$\Delta m(\text{GELT} - P)$	-0.0092 ± 0.0133	-0.0099 ± 0.0142

What each column means. The norm^2 fraction outside the span is $\|r\|^2/\|O_{\text{GELT}}\|^2$ in the metric (6) — the squared sine of the angle between the learned operator and the classical subspace. “12.8% outside” says that writing the learned operator as a combination of the seven classical operators as well as least squares allows leaves 12.8% of its squared length unaccounted

for. It involves no fit and no mass, and — crucially for Section 5.5 — it says *where the operator points*, not whether it is any good.

Z_r/Z_G is not a norm. It is the share of the *ground-state amplitude* carried by the residual, read off the cross-correlator at the fixed reference mass (Section 2.4), so it carries a sign and it asks a physical rather than a geometric question: of the operator’s coupling to the state we want, how much comes from the part the classical basis cannot express? The two columns can and do disagree — that disagreement is the entire content of Section 5.5.

$\Delta A_0(\text{GELT} - P)$ is the sharp statement, and the one to quote. It is not “GELT beats a classical operator”; it is “deleting from GELT everything the classical span can express costs this much ground-state weight”. Since P is by construction the best classical approximation available, this is a *lower bound* on the advantage over the whole span. The accompanying Δm , consistent with zero on both ensembles at 0.7σ , says the two are measuring the same state — without which the A_0 comparison would be meaningless.

5.3 Two readings that came out sideways

Two expectations were pre-registered before this measurement, and reporting how they landed is more useful than reporting only the numbers.

“A smaller out-of-span fraction than the published 12.9%/11.7%” — not as literally written. Against **deep**, the seven-level network reads 12.7% and 12.8%: the same number. At face value that is a strengthened span absorbing nothing, which would be a serious problem for the whole study. It is not, because the literal comparison mixes two different networks: the published 12.9%/11.7% is the *four-level* network against the *four-level* span. The apples-to-apples question — does a stronger span absorb the learned content? — is one fixed operator against nested spans, and there the answer is unambiguous:

$$21.1\% \longrightarrow 12.8\% \longrightarrow 2.5\% \quad (4 \rightarrow 7 \rightarrow 21 \text{ operators}). \quad (10)$$

Enlarging the span from four to seven levels absorbs 40% of the learned operator’s out-of-span norm; adding loop shapes absorbs most of the rest. The coincidence of 12.7% with 12.9% is exactly that — a coincidence between two different measurements. What it does reveal, read correctly, is that the seven-level network is *further* outside the four-level span (21.1%) than the four-level network is (12.9%): richer inputs produce a more distant operator. A study that quoted only the headline fraction would have reported the same number twice and concluded nothing.

“ $\Delta A_0(\text{GELT} - P)$ positive but below +0.097” — positive, and at the bound. Measured: $+0.0965 \pm 0.0214$, against the fair fight’s $\Delta A_0(\text{GELT} - \text{deep GEVP}) = +0.097 \pm 0.021$. The two agree to the third decimal. The expectation was that they would not: P is a better classical operator than the GEVP vector, so deleting r should cost *less* than losing to the GEVP.

That they coincide is not a coincidence, and the mechanism is visible in Table 6: $A_0(P)$ lands on $A_0(\text{GEVP})$ on both ensembles — 0.884 against 0.905 on run5, 0.944 against 0.939 on ens1. The least-squares shadow of the learned operator and the variationally optimal classical operator are the same quality of interpolator. That is a non-trivial internal consistency check: the span really does have a ceiling near $A_0 \approx 0.9$, and *the ceiling is a property of the span rather than an artifact of how the GEVP picks its vector*. The same check was made at four levels in the original study (0.830 against 0.837) and it now holds at seven.

5.4 What r is made of

The residual r is orthogonal to $\text{span}\{\text{deep}\}$ by construction, but it need not be orthogonal to everything classical. Since $\text{span}\{\text{full}\} \supset \text{span}\{\text{deep}\}$ and the extra directions are exactly

the loop shapes, projecting r onto $\text{span}\{\text{full}\}$ isolates how much of the learned residual is rectangular Wilson loops:

$$\frac{\|\text{proj}_{\text{span}\{\text{full}\}} r\|^2}{\|r\|^2} = 75.8(4)\% (\text{run5}), \quad 81.9(3)\% (\text{ens1}), \quad \text{combined } 79.6 \pm 0.3\%. \quad (11)$$

This number depends on how much of **full**’s near-degenerate span one is willing to call real. At the script default the Gram of the 21-operator span is inverted with a relative cut of 10^{-12} , keeping all 21 directions and giving Eq. (11). Cutting at 10^{-4} , which keeps 10 directions, gives 55.5(7)% and 60.7(8)%.⁴ The spread is the honest error on this statement, and it is larger than the jackknife error: it is a systematic of the projection, not of the sample.

The reading, at either end: **most of what the network found outside the smearing ladder is rectangular loops it rediscovered** — between a half and four fifths — and the remainder, about 2.5% of the operator’s total norm², is content that *no* operator in the 21-member classical basis expresses. That last piece is not negligible: it still carries $\Delta A_0 = +0.021 \pm 0.009$ (2.3σ), consistent with the fair fight’s independent $\Delta A_0(\text{GELT} - \text{full}) = +0.038 \pm 0.027$.

5.5 The untrained control, and why it is the whole point

Everything above is compatible with a deflationary explanation: the L1-ball parallel transport of the GELT block reaches offsets and path-averages that no smeared plaquette contains, so *any* network of this architecture — trained or not — would sit outside the classical span. If that is the case, the norm fraction is a fact about the architecture and carries no evidence of learning at all. This was written down as the expectation, in those words, before the control ran.

Six untrained seven-level networks (init seeds 0, 1, 2 on both ensembles), evaluated through the same pipeline against the same **deep** span, with the reference mass for the amplitude split taken from the trained network on that ensemble (0.3375 on run5, 0.3689 on ens1).

Table 7: Trained against untrained, same architecture, same inputs, same span, same 400 configurations. Seed 0’s cosh fit does not converge on either ensemble ($m = 1.23 \pm 0.43$ and 0.75 ± 0.30 against the trained 0.34 and 0.37), so it is excluded on a criterion visible without looking at ΔA_0 ; both combinations are reported.

	trained (7lv)	untrained (4 converged seeds)	untrained (all 6)
norm ² outside span	0.1277 ± 0.0025	0.2371 ± 0.0551	0.4036 ± 0.1220
Z_r/Z_G	0.1459 ± 0.0071	0.0474 ± 0.0220	0.1894 ± 0.1037
r inside $\text{span}\{\text{full}\}$	0.7959 ± 0.0027	0.0028 ± 0.0010	0.0042 ± 0.0010
$\Delta A_0(\text{net} - P)$	$+0.0965 \pm 0.0214$	-0.0911 ± 0.0420	-0.0745 ± 0.1271

Per seed, $\Delta A_0(\text{net} - P) = -0.107(61)$ and $-0.064(35)$ on run5 and $-0.197(60)$ and $-0.024(45)$ on ens1 for seeds 1 and 2; seed 0 gives $+0.70 \pm 0.91$ and -0.13 ± 0.31 , i.e. no information in either direction. The per-seed out-of-span fractions are 16%, 31% and 74% (seeds 2, 1, 0), and the two ensembles agree to better than 1% relative on each seed — as they must, since the init seed *is* the network and the ensemble only supplies configurations to evaluate it on. Three conclusions follow, the first two pre-registered and the third not.

1. **Being outside the span is architectural.** Untrained networks are 16–74% outside, against the trained network’s 12.8%. Training does not push the operator away from the

⁴Both ends are quoted in `notes/fable5.1_10-09_audit.md` §8.3. That section describes the 10^{-4} setting as truncating **full** “to the 10 directions §8.1’s estimator keeps”; strictly, the 10 directions are those of the $C(0)$ Gram of the span under the same ϵ , while §8.1’s estimator keeps 9 directions of $C(t_0)$. The two are different matrices, so both counts are correct, but they are not the same 10 and 9 directions.

classical span; it moves it *closer* while making it better. The norm fraction therefore proves nothing about learning, exactly as the pre-registration said it would not, and any write-up that leads with “12.9% outside the span” as evidence of what the network learned is making an unsupported claim.

2. **The discriminating statistic is Z_r/Z_G .** Per unit of norm, the trained residual is an order of magnitude more productive:

$$\frac{Z_r/Z_G}{\sqrt{\|r\|^2/\|O_{\text{GELT}}\|^2}} = \frac{0.146}{\sqrt{0.128}} = 0.41 \quad \text{trained}, \quad \frac{0.047}{\sqrt{0.237}} = 0.10 \quad \text{untrained}. \quad (12)$$

An untrained network has a large out-of-span component that couples to almost nothing; training builds a smaller one that couples strongly.

3. **The sign of $\Delta A_0(\text{net} - P)$ flips.** For the trained network, deleting the out-of-span part costs the whole advantage, $+0.097 \pm 0.021$. For an untrained one, deleting it *helps*: its own projection onto the classical span is the better operator, by 0.091 ± 0.042 . This is the cleanest separation in the report, because it is a change of sign rather than of magnitude, and no normalisation convention can produce it.

The third conclusion was not anticipated anywhere and is arguably the strongest of the three: **the residual is loop-shaped only when the network is trained.** 80% of the trained r is reachable inside $\text{span}\{\text{full}\}$; 0.3% of the untrained r is. An untrained equivariant network sits outside the classical span in a direction that is outside *everything* classical — some accident of the random transport average, orthogonal to the space rectangular Wilson loops span. Training rotates that content onto the part of operator space the loops do span, and then keeps a small piece that is not. That is a much more specific description of what training does than “it improves A_0 ”.

5.6 The mechanism: constructive interference, not a better operator

The naive reading of Section 5.2 — “the network found a better operator hiding outside the basis” — is wrong, and the data say so directly. On its own, r is a *poor* operator: $A_0(r) = 0.46$ and 0.42 on the two ensembles, against the full learned operator’s 0.95 and 1.07 (Table 6); the published four-level study read 0.43 and 0.37 . It does not win by being good. It wins by adding *coherently*.

The arithmetic is the reason the amplitude split is taken at one fixed mass. With $Z_G = Z_P + Z_r$ exactly, removing r removes a fraction

$$1 - \left(1 - \frac{Z_r}{Z_G}\right)^2 \quad (13)$$

of the ground-state *intensity*. For the published four-level decomposition $Z_r/Z_G = 0.131$, so Eq. (13) gives 24.5% of the intensity removed by deleting 12.9% of the norm²; against **deep**, $Z_r/Z_G = 0.146$ gives 27% from 12.8% of the norm². In both cases the amplitude share is about twice what the norm share would suggest, the ground-state fraction $A_0 = A/C(0)$ falls accordingly, and that is the whole effect. (The numbers are of the right size to account for the observed drops — $0.903 \rightarrow 0.830$ at four levels, $0.955 \rightarrow 0.884$ at seven — but are not expected to match to three digits, because each arm’s A_0 is quoted from its own cosh fit rather than from the fixed-mass amplitudes.)

Stated carefully for the thesis: what the network found is a *direction* that is mediocre alone and valuable in superposition — precisely what a variational method exists to exploit, and precisely what a basis of four, seven or twenty-one hand-built operators cannot reach unless somebody thinks of the right operator to add.

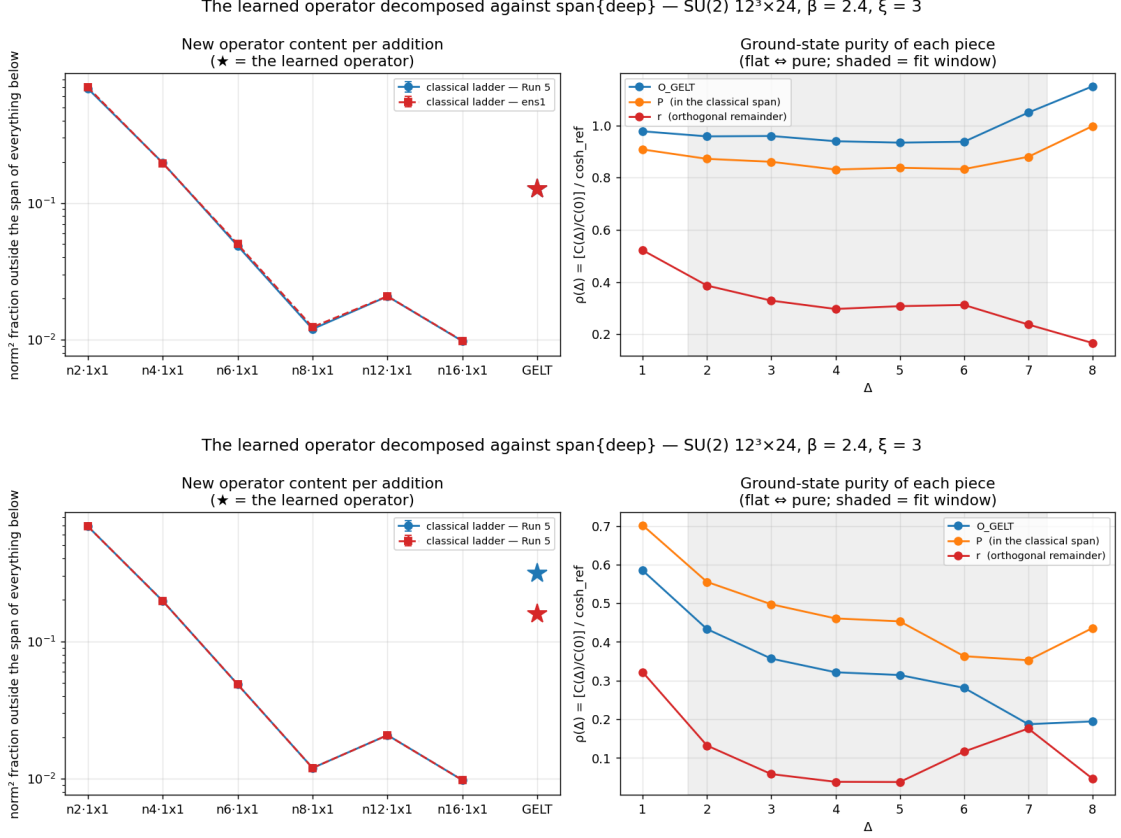


Figure 3: The decomposition against span{deep}. **Top:** the trained seven-level network. Left, the ladder increments of Section 5.7 with the learned operator as a star; right, $\rho(\Delta) = [C(\Delta)/C(0)] / \cosh_{\text{ref}}(\Delta)$ for the three pieces — flat means pure, and the excess at small Δ is excited-state contamination. O_{GELT} (blue) sits above its own classical shadow P (orange) across the whole fit window; r (red) is flat from $\Delta \approx 3$, i.e. it couples to the same ground state rather than dying with the UV. **Bottom:** two untrained networks (seeds 1 and 2; the left panel’s two stars are their out-of-span fractions, 31% and 16%). The ordering in the right panel is *inverted* — the projection P lies above the network’s own operator — which is the sign flip of Section 5.5 drawn rather than tabulated. (The bottom-left legend labels both traces “Run 5”; the classical ladder is in fact ensemble-specific and the two curves coincide to line width, which is the cosmetic reason they are indistinguishable.)

5.7 The scale: the classical ladder’s own increments

The remaining question from Section 2.4 is what 13% means. The calibration is the same statistic applied to the classical operators themselves: how much new content does each rung of the smearing ladder add to the span of the rungs below it?

addition	run5	ens1
APE $\times 2$ outside span $\{ \times 0 \}$	0.6891 ± 0.0098	0.7049 ± 0.0084
APE $\times 4$ outside span $\{ \times 0, 2 \}$	0.1971 ± 0.0047	0.1957 ± 0.0043
APE $\times 6$ outside span $\{ \times 0, 2, 4 \}$	0.0486 ± 0.0014	0.0500 ± 0.0012
APE $\times 8$ outside span $\{ \times 0, 2, 4, 6 \}$	0.0120 ± 0.0004	0.0124 ± 0.0003
APE $\times 12$ outside span $\{ \times 0, \dots, 8 \}$	0.0207 ± 0.0006	0.0208 ± 0.0006
APE $\times 16$ outside span $\{ \times 0, \dots, 12 \}$	0.0098 ± 0.0003	0.0098 ± 0.0003
GELT outside span{the whole 7-level ladder}	0.1273 ± 0.0037	0.1279 ± 0.0033

Each number replicates across two independently sampled ensembles to the third decimal —

these are properties of the operator basis, not of the sample. The first three rungs fall by a factor of 3.5–4 each, and the learned operator’s 12.8% exceeds every increment past the second: a seventh smearing level buys 1% of new content, the network buys 13%.

One extrapolation is withdrawn. The original study fitted the decay of the first three rungs and extrapolated that the *entire* remaining tail of the ladder — rungs 8, 12, 16 and beyond — was worth $\approx 2\%$ of new content, concluding that “the classical smearing ladder is a geometric series that has essentially converged”. The three rungs have since been measured directly and they are 1.20%, 2.07%, 0.98%: a tail worth $\approx 4\%$, not $\approx 2\%$, and **not monotone in the rung index**. The non-monotonicity is not noise — the errors are at the fourth decimal and both ensembles show the same rise at $\times 12$ — it is that the ladder’s own steps are $\Delta 2, \Delta 4, \Delta 4$, so the rungs are not equally spaced and the series was never going to be geometric in the rung index. The extrapolation is withdrawn; the qualitative reading (the ladder has slowed, the network is not on that curve) survives, but “the ladder has converged” is too strong. It has slowed, and its tail still matters at the percent level that this whole study works in.

The same arithmetic underwrote a prediction the decomposition made about the fair fight before that fight was readable: **deep** could not close a 13% gap, because more smearing is the direction the ladder has already converged along, and only real loop geometry could move A_0 materially. Both halves held — **deep** leaves $+0.095 \pm 0.021$, and **full** is where the movement is — but the numbers behind the prediction (rungs 8, 12, 16 at 1.3%, 0.35%, 0.1%) were too optimistic by the factors above. The conclusion was right for a partly wrong reason, and both halves of that should be said.

6 Honesty section: what these three studies do not settle

The pre-registration, and what it bought. The observable, the estimator family, the fit conventions, the measurement grid and the three readings with their tests were committed to git before any of the numbers in Sections 3–5 existed. What that bought is narrow: it removes the objection “the opponent was chosen after the fact”. It does *not* make the results stronger than their error bars, and it did not stop two of the three readings from failing their own tests. Two limits apply. The estimator rule had to be amended (Section 3.2); the amendment is recorded with its reason and the four-setting table shows it moves nothing, but an amended rule is weaker than an unamended one. And the decomposition study was originally written *after* its own first measurement — its window and metric were inherited rather than chosen, both are scanned rather than reported at one setting, and the two ensembles were analysed with identical code and agree, none of which is as good as pre-registration and should not be described as if it were. The two *controls* in that section *were* pre-registered, before they ran.

The seven-level network is a reactive point. It was trained *after* the **deep** arm was seen to win on the run5 ensemble. The classical arms themselves were named before the fair fight ran and that pedigree is real, but the network at the right-hand end of the main chain was a response to a number. The thesis must say so where it quotes the seven-level numbers. Everything else on the curve — the thin retrain, the width control, the random trace, the estimator and the readings — was fixed before its numbers existed.

Two expectations came out sideways , and both are in Section 5.3 rather than smoothed over: the out-of-span fraction against the strengthened span did not shrink as literally predicted (it is the same number, for a reason that turns out to be about comparing two different networks), and $\Delta A_0(\text{GELT} - P)$ landed *at* the bound rather than below it (for a reason that is an internal consistency check in disguise). A write-up that reported only the numbers would have lost both.

Saturation, a compressed observable, and $A_0 > 1$. $A_0 \leq 1$ and every strong arm reads 0.90–1.07. The 7lv point is excluded from the curve’s tests on that basis, and — less comfortably — the whole of Section 3.3 is read off arms inside the saturation band, so “loop shapes close about 60% of the gap” is a ratio of two small differences near a ceiling: the right qualitative reading, and not a number to quote with an error bar. Relatedly, the learned operator reads $A_0 = 1.013 \pm 0.062$ at four levels and 1.066 ± 0.059 at seven on ens1. That is a fit artifact of a nearly pure operator, not a probability violation (Section 2.1); it is inherited from the published table, not made worse here, and reported unclipped so that $1 - A_0$ keeps its sign in every difference. The thesis should say so once rather than leave a reader to notice it.

The contact-term test does not survive a longer ladder. The original study repeated the whole decomposition under the metrics $C_{ab}(\tau)$ for $\tau = 1, 2$, which weight the light states more heavily: a contact term — UV content living in $C(0)$ and gone by $\Delta = 2$ — must *shrink* as τ grows. On the four-operator span it *grew* ($0.129 \rightarrow 0.156 \rightarrow 0.197$ on run5), which is the opposite of a contact term and is corroborated by $\rho(\Delta)$ for r being flat from $\Delta \approx 3$ (Figure 3, top right). That test does not transfer. On $\text{span}\{\text{deep}\}$ the $C(2)$ Gram has $\lambda_{\min} = -2.9 \times 10^{-5}$ and on $\text{span}\{\text{full}\}$ it is -0.73 : $C(\tau)$ is then not an inner product on those spans, the “projection” is undefined, and the jackknife error duly explodes (to ± 383 on run5). The script now checks the spectrum and prints NOT USABLE beside the row rather than a number. **The contact-term conclusion stands only where it was measured** — the four-operator span, where $C(1)$ and $C(2)$ are positive definite — and is not restated for the longer ladders.

The shape-span fraction depends on the projection cut. 80% at $--\text{proj-eps} = 10^{-12}$ and 58% at 10^{-4} (Section 5.4). The 21-operator span’s $C(0)$ Gram has condition number 6.9×10^8 , so a cut is unavoidable; which value is “right” is a judgement about how much of a near-degenerate span is real. Both ends are quoted everywhere the number appears. Nothing in the **deep** decomposition moves with this cut ($0.1273 \rightarrow 0.1276$ and $+0.0712 \rightarrow +0.0717$ from 10^{-12} to 10^{-4}); only the shape fraction does.

One untrained seed is excluded. Seed 0’s cosh fit does not converge on either ensemble ($m = 1.23 \pm 0.43$ and 0.75 ± 0.30 against the trained 0.34 and 0.37; $A_0 = 1.02 \pm 0.89$ and 0.37 ± 0.25). It is excluded on a criterion — “the fit did not converge” — visible without looking at ΔA_0 , which is the only kind of exclusion that is safe here, and *both* combinations appear in Table 7. Including it does not change the sign of $\Delta A_0(\text{net} - P)$; it inflates the error threefold, which is what an unconverged fit should do.

The right-hand end of the curve is not matched, and one point is off width. The shape-input network was never trained, so the 7lv+3sh point compares a GELT *without* loop shapes to a GEVP *with* them; the figure draws it as an off-chain classical marker with no trained counterpart, and the fair fight’s $\Delta A_0(\text{full}) = +0.038 \pm 0.027$ is the number that reading rests on (closing it is Section 7, item 5). Separately, the untrained trace at 4lv is at $d_{\text{model}} = 16$ rather than the curve’s 24; six eval-only runs would fix it, and no reading depends on it, since that trace is far below the classical arm at every x and the width control (Section 4.4) finds the two widths indistinguishable.

One theory, one lattice family, one architecture. $SU(2)$, $12^3 \times 24$, $\beta = 2.4$, $\xi = 3$, receptive radius $R = 2$, the Run-5 hyperparameters, the 0^{++} channel. There is no continuum limit here and none is claimed. Every statement in this report is about *this* learned operator against *these* classical bases on *these* two ensembles.

7 What to do next, and why

The items below are ranked by value per unit of cost, and each is stated as what it measures, why the results above make it the right next thing, what it costs, and what it would license.

1. The excited-state gate (laptop, minutes, offline). Every statement in this report is compressed against the $A_0 \leq 1$ ceiling (Section 6), and the only way past it is a harder observable — the obvious candidate being the first excited state, with the network emitting a *vector* of operators and a GEVP run on the learned pair. Before any training code is written, one question must be answered: does a classical comparator for such a basis exist on these ensembles at all? That is a pass/fail test costing minutes, offline, from the tracked `dumps/su2_fair_fight_obars_run5.pt`: solve the GEVP over $\text{span}\{\text{deep}\}$, read the *second* eigenvalue’s effective mass at $\Delta = 1$ and 2 under a delete-block jackknife, and pass only if the two agree within 1σ with the $\Delta = 2$ error below 25% on at least one ensemble. If it fails the option closes for free; if it passes, the design is already fixed (`mlp_out = 2`, loss $-\text{Tr}[(C(0) + \varepsilon)^{-1}C(1)]$ over the learned 2×2 block, which by Ky Fan is maximised at $e^{-m_1} + e^{-m_2}$ and penalises collapse onto two copies because $C(0)$ goes singular). The cheapest item in the plan, and the only one gating a training run.

2. Write the thesis text (no compute). Every number the spectroscopy chapter was waiting on now exists. Concretely: the single “ $+0.078 \pm 0.022$ ” sentence becomes the curve statement of Section 4.3, with the four-level number kept as the first result and the input-matched $+0.095 \pm 0.021$ as the headline; the overlap table becomes the per- x table; and the fair fight and the decomposition, which the thesis does not currently mention at all, get a paragraph each. The decomposition paragraph can now say *learned* — but on Z_r/Z_G and the sign flip of Section 5.5, never on the norm fraction, which the untrained control shows to be architectural. This is the highest-value item that costs nothing but writing, and doing it before the next measurement is what keeps the measurements honest.

3. Fix the $\mathbb{Z}_2$ smearing (≈ 12 h GPU). The one live defect in the repository. At the production $\alpha = \frac{1}{2}$ the APE update for two spatial staples is $V = (1 - \alpha)U + (\alpha/2)(s_1 + s_2)$, which is exactly 0 whenever the two staples disagree with the link — and $\mathbb{Z}_2$ ’s projection sends $0 \rightarrow +1$, a value that does not transform under a gauge transformation. Measured on production configurations the ladder freezes after one step, 99.9% of the changed links are gauge-dependent tie-breaks, and the timeslice operator moves by 1.82σ under a gauge transformation. The consequence for the tables in the attention report is that the $\mathbb{Z}_2$ classical basis is *one* operator and not four (the dumps already record `gevp_fell_back=True`, `n_ops=1`), and the $\mathbb{Z}_2$ networks were trained on four channels of which three are byte-identical. The fix is unprojected fat links — linear in the links, hence exactly covariant — with the cheap first step being to evaluate the existing checkpoints on covariant inputs before retraining anything. $SU(2)$ is verified clean at 1.4×10^{-15} , so nothing in *this* report is affected; the next item, however, depends on it.

4. Dump per-configuration $\bar{O}$ from the $\mathbb{Z}_2$ attention script (small code change). The tooling built for Section 5 — selectable spans, the shape-span projection, the fixed reference mass, the untrained control — is generic: it takes per-config $\bar{O}$ arrays and a classical basis, and the attention field is exactly such an operator. Dumping those arrays transports the whole analysis to the attention field with no GPU re-run, turning “the attention field has a mass and a good overlap” into “here is the part of it the classical basis cannot express”. The catch is item 3: until the $\mathbb{Z}_2$ smearing is fixed that classical span is *one-dimensional*, and a decomposition against a one-dimensional span is not worth running. Sequence the two together, in this order.

5. Train a shape-input network (2 trainings). Optional before Section 5.4, this is now the direct test of its main finding. If 80% of what the network found outside the smearing ladder is rectangular loops it rediscovered, handing it loop shapes as input channels should let it spend its capacity elsewhere — and the prediction is sharp: the out-of-span fraction against $\text{span}\{\text{full}\}$ need not simply drop, since the network is now *given* the directions it previously had to build. It also closes the unmatched right-hand end of the curve (Section 6). Cost: cubic-symmetrised matrix-valued $R \times T$ loops as extra adjoint channels, about a day of code, then two trainings of roughly eight hours each.

6. The $SU(2)$ β -scan of the attention field (≈ 6 h GPU, zero code). The attention report’s $SU(2)$ row is a single coupling, because $\beta = 2.4$ is the only one with a trained network, and the script refuses to print aggregates below three couplings rather than emit a meaningless correlation. Sampling four more ensembles and evaluating the existing network off-diagonal turns that row into a scan, making the ξ_A claim in $SU(2)$ the same kind of statement it already is in $\mathbb{Z}_2$. Pure GPU filler with no code to write — and it must not share the GPU with a training, which is the only reason it is not higher.

7. The matched-parameter L-CNN (2 trainings, plus a model switch). The one baseline the thesis names and has never run. The argument for attention currently rests on beating a *non*-equivariant CNN, which is a weak opponent for the same reason **published** was: it is not the thing a practitioner would build. `gelt/lcnn.py` already mirrors GELT’s I/O, so the change is a model switch and axis-aligned transports in place of the L1-ball average. The right presentation is not a standalone number but a *fourth trace* on Figure 2: the same $A_0(x)$ for a matched-parameter L-CNN. The pre-registered expectation is parity with GELT — the smeared inputs hand both networks the staple content, and at matched depth the loop degree is the same — and parity is the honest form of “attention costs nothing, and the attention field is a measurement a convolution cannot produce”.

8 Conclusions

One number, $\Delta A_0 = +0.078 \pm 0.022$, has been replaced by three answers to three different questions, all measured offline on the same held-out configurations under conventions fixed in advance.

- **The comparator was under-powered, and strengthening it does not overturn the claim.** Against the input-matched seven-level ladder the learned operator leads by $+0.095 \pm 0.021$; against the strongest 21-operator basis the repository builds, by $+0.038 \pm 0.027$. Loop shapes — not more smearing — close most of the gap, as the decomposition predicted before that arm was readable. Reading it at all required truncating the whitening rather than flooring it, and the pre-registered selection rule had to be amended, because two arms fail a gate the GEVP at $t_0 = 1$ does not guarantee: the projection maximises $C(2)/C(1)$ while the gate tests $C(2)/C(0)$.
- **The advantage is training, not architecture, and not merely richer inputs.** At every input depth the observable can resolve, one learned operator carries more ground-state weight than the optimal linear combination of the *same* inputs: $+0.252 \pm 0.056$ on thin links, $+0.075 \pm 0.019$ at four smearing levels. An untrained network of identical architecture sits *below* the classical GEVP at every point, and a width control shows $16 \rightarrow 24$ channels changes nothing. None of the three pre-registered readings survives its own test unmodified; the honest fourth statement is that the gap is largest where the classical basis is weakest and decays toward the saturation ceiling.

- **What the network found is a mediocre direction that is valuable in superposition, and it is mostly loop shapes.** The same operator lies 21.1%, 12.8% and 2.5% outside the 4-, 7- and 21-dimensional classical spans; deleting the out-of-span part costs the entire advantage ($+0.097 \pm 0.021$ against the strong span, at unchanged mass); 80% of it is rectangular Wilson loops the network rediscovered, leaving $\approx 2.5\%$ of the operator’s norm that no member of a 21-operator basis expresses and that still carries $+0.021 \pm 0.009$. Alone the residual is a poor operator ($A_0 \approx 0.42\text{--}0.46$): it wins by constructive interference, carrying about twice the amplitude share its norm share would suggest.
- **The control decides which of those statistics means “learned”.** An untrained network is 16–74% outside the same span — further out than the trained one — so the norm fraction is architectural and carries no evidence of learning. Training moves the operator *closer* to the classical span while making it better. What separates learned from architectural is the amplitude share ($Z_r/Z_G = 0.146$ against 0.047, or 0.41 against 0.10 per unit of norm), the fact that the residual becomes loop-shaped at all (80% against 0.3% inside the richer span), and above all the sign of $\Delta A_0(\text{net} - P)$, which is $+0.097 \pm 0.021$ trained and -0.091 ± 0.042 untrained: for a trained network, deleting the out-of-span content costs the advantage; for an untrained one, deleting it helps.

What remains is a ceiling rather than a missing measurement. Every strong arm on these ensembles reads $A_0 \geq 0.90$ against a bound of 1, so the differences that separate the methods are being measured in the last tenth of the available range, and no number of configurations changes that. The next real step is a harder observable — the first excited state, gated by a test that costs minutes — rather than a better estimator for this one.

Reproducing the numbers. All three studies are offline, CPU, and read only files tracked in `dumps/`. The figures live under `results/`, which is gitignored but regenerable in minutes: Figure 1 from `SFF_TRUNCATE=1 SFF_BASES=1 python scripts/su2_fair_fight.py <dump>`, Figure 2 from `python scripts/input_architecture_curve.py` (which also writes Table 4 as a paste-ready `.tex` fragment), and both panels of Figure 3 from `python scripts/operator_decomposition.py` with `--basis=<obars>:deep`, `--shape-span=<obars>:full` and, for the untrained arm, `--m-ref=<trained mass>`. The source of truth for every number quoted here is `notes/fable5.1_10-09_audit.md` (§1–§2 pre-registration, §8.1 estimator and shape verdict, §8.2 curve, §8.3 decomposition and controls) and `notes/operator_decomposition.md` (§2 theory, §4 published results, §5 controls, §6.1 the withdrawn extrapolation, §7 limitations).